
Groundwater drought assessment in Southwestern Bangladesh

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Abstract: Globally, groundwater level depletion (groundwater drought) is a major issue. This issue is spreading throughout Bangladesh and poses an ongoing threat to safe water access. To comprehend the groundwater drought scenario in Bangladesh's salinity-prone southwestern coastal region, we employed a threshold level technique. The drought scales are determined using three threshold levels (170%, 190%, and 195% of mean water level). Findings indicate that the region has experienced moderate to severe drought in the past. Moderate drought periods often last two to three months. It is anticipated that occurrences of groundwater drought would advance the current salinity front further upstream.

Keywords: groundwater; drought; Bangladesh; coastal Bangladesh; groundwater drought index; threshold level; water deficit.

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1 Introduction

Groundwater is one of the most valuable water resources. Depletion of this valuable resource is a collective concern, and its effect is more prominent in developing countries (Hasan et al., 2018a). Though it is one of the major sources of water supply in the country (Haque, 2018). Groundwater governance is overlooked in Bangladesh until the 1990's (Bhattacharjee et al., 2019). It is the key source of irrigation for crop production (Dey et al., 2017; Zahid et al., 2018) and provides almost all of the drinking water supply (Shamsudduha, 2018). This safe water source is now short in supply due to overexploitation, lack of recharge, climate change effects, and degrading water quality from several sources of contamination. To minimise this shortage and ensure sustainable water resource development, research on the state of groundwater is obligatory. While there are limited studies regarding groundwater deficiency (potentiality) in Bangladesh, these studies have been confined mainly to the northwestern part of the country (e.g., Acharjee et al., 2017; Adhikary et al., 2013; Dey et al., 2012; Ferozur et al., 2019; Hasan et al., 2018b; Islam et al., 2016, 2017; Mustafa et al., 2017; Rahman et al., 2017; Shahid et al., 2015). Because it is prominently established that the northwestern region is already facing an acute shortage of irrigation and drinking water supply (Rahman et al., 2018). Therefore, the southwestern coastal region remained inattentive despite its vulnerable location and relevant importance. Although, groundwater resources in this region are at the greatest risk now (Zahid et al., 2018). In southwestern districts, groundwater is characterised by a high concentration of saline water due to various human-made and natural effects (Shammi et al., 2017, 2016). Besides, the water table is reported declining (Shamsudduha et al., 2009). Consequently, this region is one of the most vulnerable areas in terms of access to safe drinking water in the country (Hasan et al., 2018b; Zahid et al., 2018).

A study on safe water sources reveals that only 12% of the total area in Satkhira (a southwestern coastal district) is suitable for installing tubewells (Saha et al., 2018). Nowadays, climate change is triggering the issue. Rising sea levels caused by climate change would accelerate salinity intrusion (Haque, 2018). Also, the temperature has been increasing along with a changing rainfall pattern (Rakib, 2018). This change has a direct relation to groundwater drought (Mustafa et al., 2017). The occurrence of groundwater drought will cause a freshwater shortage in drought periods. On the other hand, this will facilitate an extended salinity front from the Bay of Bengal. The salinisation from underground sea streams will pollute existing freshwater (Abedin et al., 2014). Outcomes will be enhanced inland groundwater and surface water salinity even in wet periods. To cope with this hindering phenomenon and find suitable mitigation and or remediation strategies, continuous monitoring and analysis of groundwater level are vital. It will facilitate the detection of factors that need to be managed to ensure sustainable planning because modelling and prediction of groundwater fluctuations are the initial stage of effective management (Sakizadeh et al., 2019). Therefore, an attempt is made in this

study to understand the groundwater drought scenario in the southwestern coastal region of Bangladesh using time series data.

Drought characterisation using time series data is the most frequently applied technique (van Lanen and Peters, 2000). Among several available approaches, the threshold level method (Tallaksen et al., 1997; van Lanen and Peters, 2000; Yevjevich, 1967) using time series data is the most commonly used (Tallaksen and van Lanen, 2004). It can be applied to any time series, irrespective of the characteristics of the time series (Peters et al., 2006). Hence, we used the threshold level method. The spatial distribution of drought magnitude is plotted to identify the spatial severity of the groundwater drought in the study area. A normalised drought index is used to identify the most vulnerable region.

2 Materials and methods

The threshold level approach is used to assess groundwater drought in the southwestern region (Figure 1) of Bangladesh. Weekly groundwater time series data (1998–2019) were collected from Bangladesh Water Development Board (BWDB). As drought has been calculated on a monthly basis, weekly data were converted to a monthly average. Missing values were discarded from the analysis.

As the groundwater level is a state variable, the ‘deficit volume’ calculated with the threshold level approach should be interpreted as the cumulative departure (CD) from the threshold. This gives the best result in understanding groundwater drought (Peters and van Lanen, 2000; van Lanen and Peters, 2000). Therefore, in this study, CD from a threshold level is used to identify groundwater drought. This is the summation of groundwater level departed below a threshold level. Mathematically this can be written as:

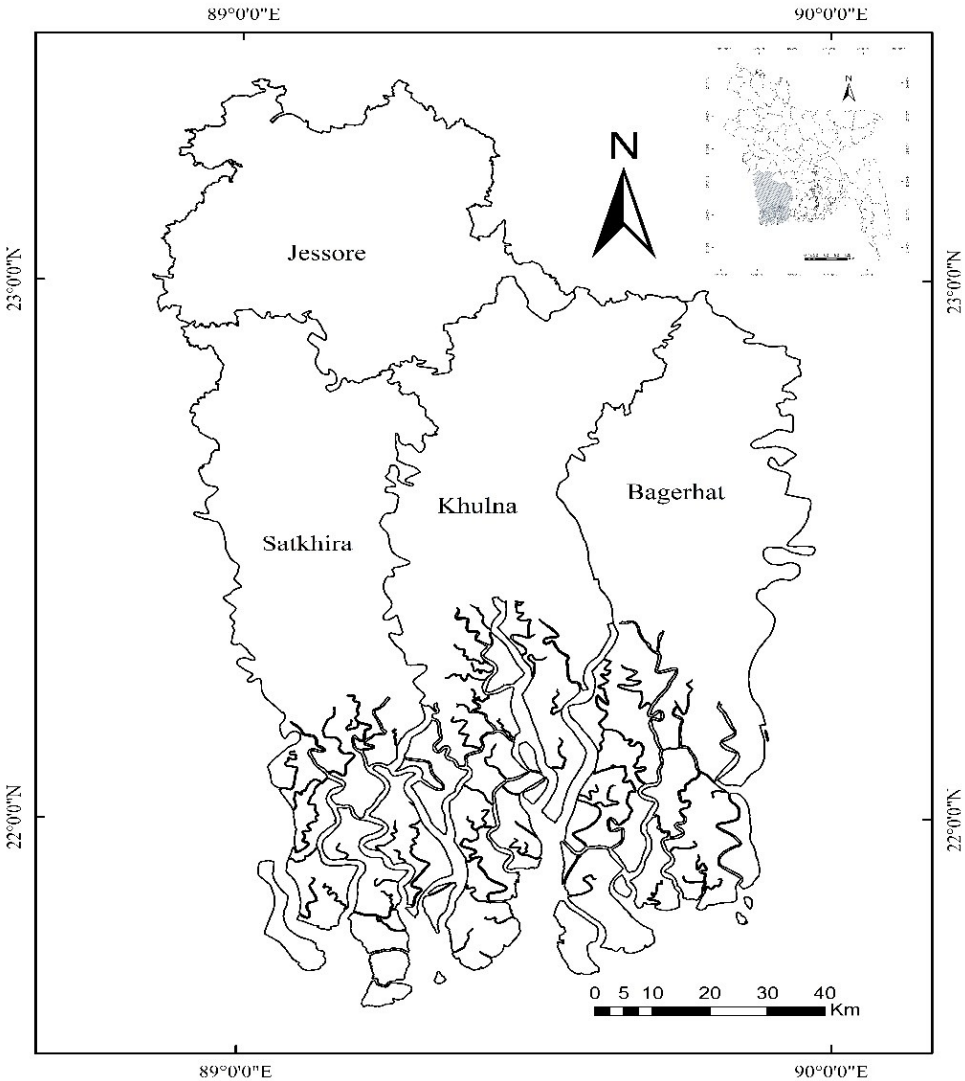
$$CD_i = CD_{i-1} + \begin{cases} \sigma_i & \text{if positive} \\ 0 & \text{if otherwise} \end{cases} \quad \text{where } \sigma_i = \text{data}_i - \text{threshold level}$$

The normalised departure can be found by the following equation:

$$CD_{\text{normalised}} = \frac{\sum CD_i}{n} \quad \text{where } n \text{ is the count of } \sigma, \text{ when } \sigma \text{ is positive}$$

Water deficit is normalised to smooth the effects of missing values. The normalisation of CD is useful to identify vulnerable areas especially if there are missing data or extreme values. Nevertheless, it can be seen from the equation that the chosen threshold has a large influence on the identified severity of drought. Therefore, the determination of a suitable threshold is an important step (Peters et al., 2003). We have selected three threshold levels, viz., 170% of the mean, 190% of the mean, and 195% of the mean water level to illustrate the severity of groundwater drought at each site. The duration of a drought event is calculated from the time of consecutive droughts. The spatial map of groundwater drought is produced using the inverse distant weighting (IDW) technique.

Figure 1 Location of the study area in Bangladesh

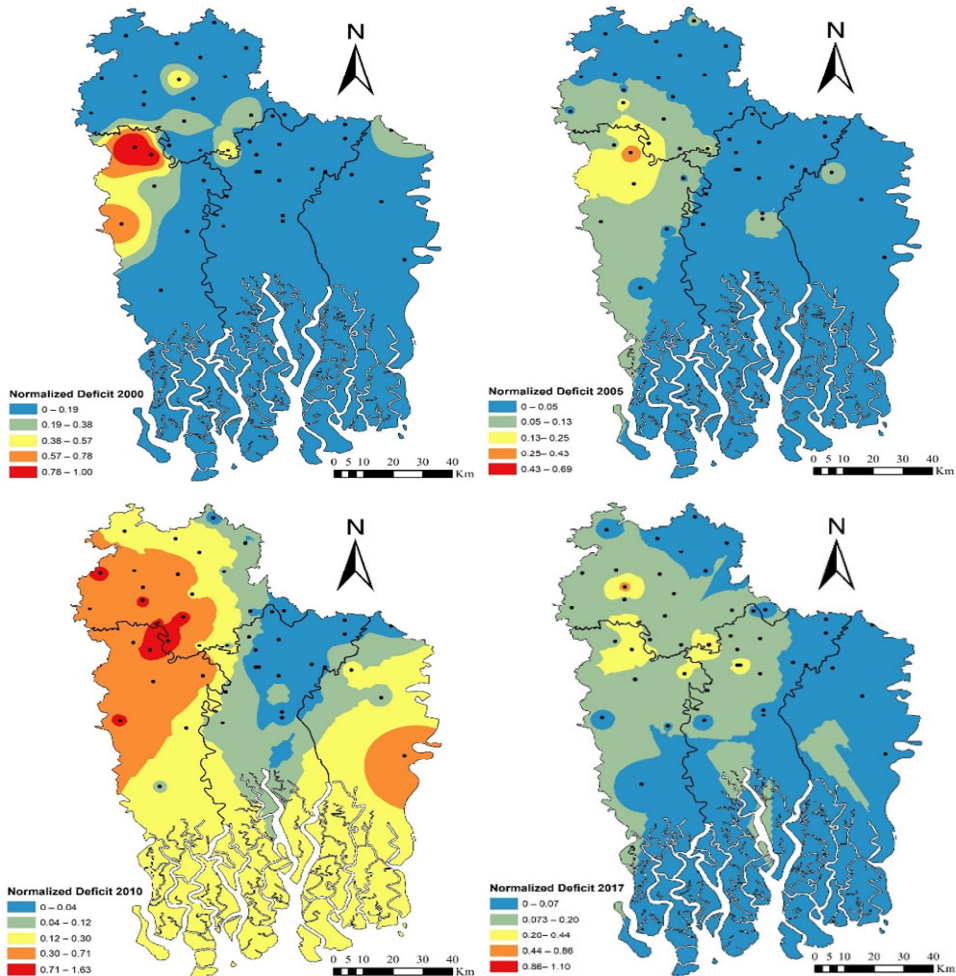


3 Results and discussion

Groundwater drought is quantified using a cumulative threshold approach in this study. As the time span increases, the cumulative deficit also increases and becomes a gigantic value. Therefore, the total deficit of a single station is normalised, dividing the deficit value by the number of months when the drought occurred. Using a normalised value smoothes the fluctuation of yearly drought and cannot be used for detecting the duration of drought events (starting or ending) and the magnitude of deficit volume. However, normalised total drought can give information about the most vulnerable zones – a useful feature for the planning of groundwater management.

To understand the spatial extent of drought in the study area, the annual normalised deficit for the threshold of 170%, 190%, and 195% of the mean groundwater level are presented in Figures 2, 3, and 4. Also, the total normalised water deficit for the study period is presented in Figure 5. The spatial distribution of normalised annual deficit for a threshold of 170% of the mean groundwater level for the years 2000, 2005, 2010, and 2017 (Figure 2) shows that the northern part of Satkhira District is the most vulnerable to drought.

Figure 2 Spatial distribution of normalised annual deficit for a threshold of 170% of the mean groundwater level (see online version for colours)

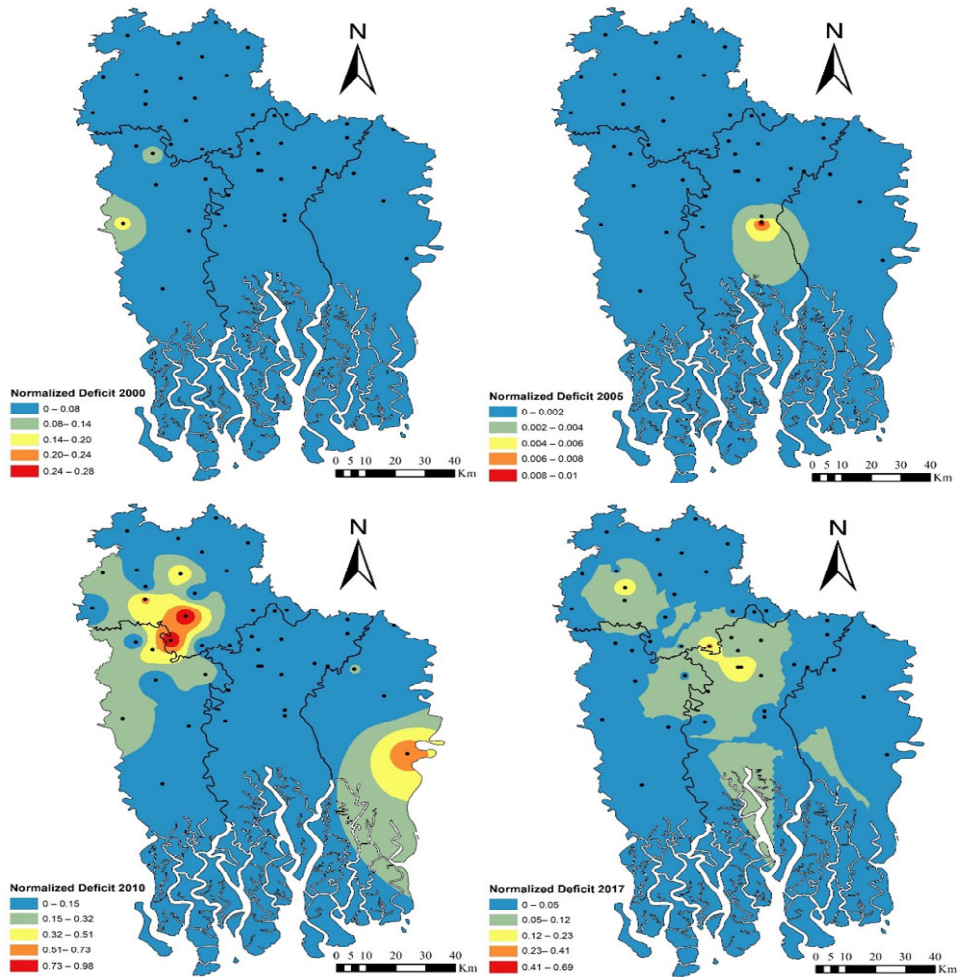


It can be stated that Kalaroa Upozilla of Satkhira District is the most vulnerable in terms of groundwater drought in the whole study area. Jessore is found moderately vulnerable, especially in the year 2010. However, it is observed that the spatial extent of drought varies year to year, as seen in the figure.

To understand the extreme vulnerability spatial distribution of annual normalised deficit for a threshold of 190% of the mean groundwater level is presented in Figure 3.

Interestingly, the spatial extent of extreme vulnerability differs from the results in the 170% of mean level. Maps of 2000, 2005, 2010, and 2017 show a spatial variation of vulnerability in the study area. However, the variation does not include the previous status of the Satkhira District. Further increment of threshold level to 195% of the mean value shows that normalised annual deficit reduces below unity. Still, there is interannual variation in the deficit value. For example, there is no extreme drought identified for the year 2005. However, there are drought events in other years. The spatial extend maps demonstrated that the severity of this magnitude of drought varies spatiotemporally (Figure 4).

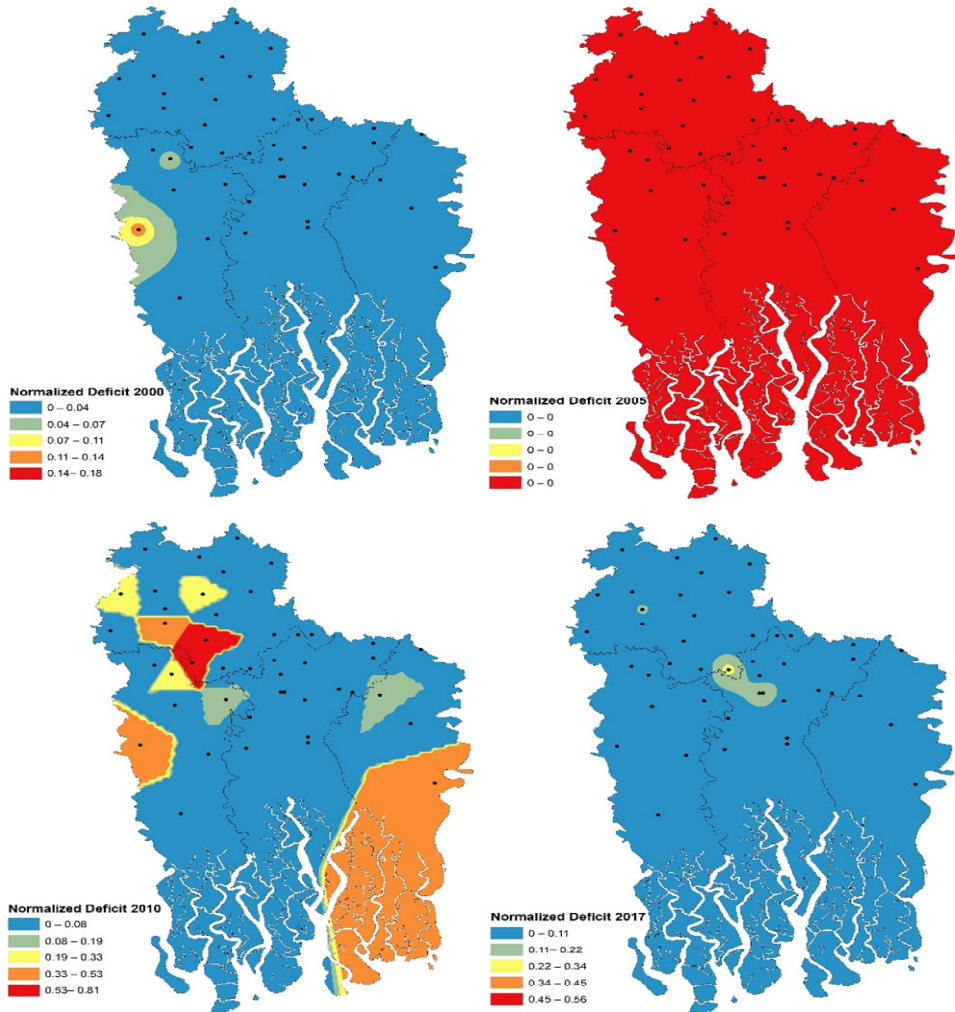
Figure 3 Spatial distribution of normalised annual deficit for a threshold of 190% of the mean groundwater level (see online version for colours)



As the spatial extent of the annual water deficit varies year to year, the total normalised water deficit is further presented in Figure 5. Normalisation smoothes the year-to-year variation and might help in efficiently discovering the most drought-prone areas in the study area. The figure illustrates that parts of the Satkhira and Jessore Districts are the

most vulnerable to groundwater drought. More explicitly, Satkhira Sadar and Kalaroa Upozilla are the most vulnerable compared to the other parts of the Satkhira District. Normalised water deficit is found greater than 2 m for several wells in the Kalaroa Upozilla. On the contrary, Jessore District has a normalised water deficit ranging from 1 m to 1.6 m.

Figure 4 Spatial distribution of normalised annual deficit for a threshold of 195% of the mean groundwater level (see online version for colours)



While normalised water deficit gives an idea about the identification of vulnerable zones, a time series plot of annual water deficit will help to understand the onset, severity, and duration. Considering these, the time series plot for an example well is presented for the most vulnerable Kalaroa Upazilla. The time series plot of the well (ID GT 8743004) is presented in Figure 6. This well is chosen for several reasons. There are comparatively fewer missing values in this well. Also, a result of normalised total and annual deficit shows that drought is far more extreme in the vicinity of this well.

Figure 5 Normalised total water deficit for the study area (see online version for colours)

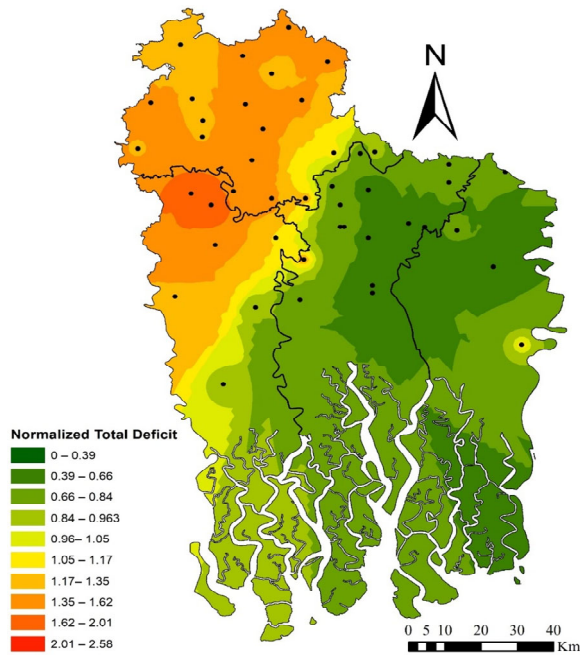
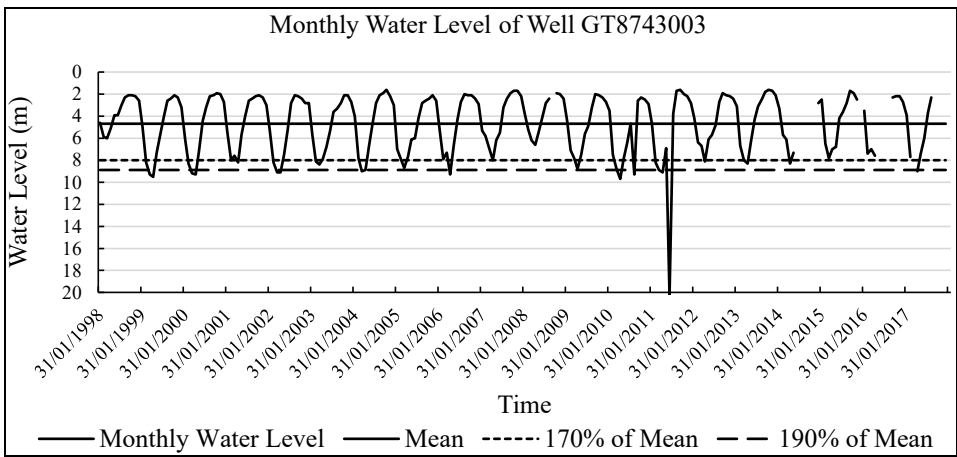


Figure 6 Definition sketch of an example well



The time series plot of the monthly water level (Figure 6) shows that there is an outlier in the series (June 2011). This outlier affects the total deficit calculation for this particular well. However, it does not greatly affect the mean of the time series. Replacement of this outlier with mean value shows that the mean value remains the same up to one decimal point. Although outliers can affect total deficit calculation, normalisation smoothes their effect.

If we assume groundwater drought can only occur in situations when the water table is below the average (4.7 m) level, there are droughts almost every year, most ranging

from six months in duration. As the threshold level is changed to 8 (170% of the mean), drought events and duration are significantly reduced. Whether previously there was a drought almost every year, time gaps between consecutive droughts increase here. Results show that the duration of the droughts reduced to 2 to 3 months in most cases. It is also found that drought events are mainly confined to February to June. Change in the threshold level to a value of 9 (190% of the mean) shows that drought events are further reduced and have a shorter spell. It is also found that groundwater drought repeatedly coincides with the winter and early summer seasons when agricultural drought usually prevails.

4 Conclusions

This study investigates the occurrence and magnitude of groundwater drought in the southwestern region of Bangladesh by threshold level approach. Cumulative deficit (total and normalised) is measured using 170%, 190%, and 195% of the mean water level as the threshold. Normalised total and an annual deficit of groundwater level show that groundwater drought is factual for this area. Although the annual water deficit varies year to year, it has been seen from the total normalised water deficit that Satkhira Sadar and Kalaroa Upozilla are the most vulnerable compared to the other part of the study area. The total normalised water deficit is found greater than 2 m for several wells in the Kalaroa Upozilla of Satkhira District. While Jessore has a normalised total deficit of 1 m to 1.6 m which seems to tumble behind Satkhira. It has been seen that the presence of missing values or outliers in the data can affect the cumulative deficit calculation, but their effects have been smoothed using the normalised deficit value. Because normalisation reduces the effect of missing values or outliers in the series. Besides, this can also be used for identifying vulnerable areas. The time series plot of an example well exhibits that selection of threshold level has a great significance in deficit calculation. This also unveils that moderate drought (threshold of 170% of mean) is usually 2–3 months in duration. It is also found that groundwater drought repeatedly coincides with the winter and early summer seasons.

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